Problem Statement Report: Predicting Milk Consumption

1. Introduction:

The dairy industry relies heavily on accurate predictions of milk consumption to optimize production, distribution, and sales. In collaboration with Provilac Dairy, we aim to develop predictive models for estimating milk consumption in liters per day across 21 products and 4 cities. The initial focus will be on three key products: raw milk, pasteurized milk, and A2 milk.

2. Background:

Accurate predictions of milk consumption are crucial for dairy industry stakeholders to optimize production, minimize waste, and meet customer demand effectively. Historical data on milk consumption provides valuable insights into consumption patterns, seasonal variations, and long-term trends, making it an essential resource for developing predictive models.

3. Problem Statement:

The challenge is to develop predictive models capable of accurately estimating milk consumption in liters per day across 21 products and 4 cities, with a particular focus on raw milk, pasteurized milk, and A2 milk. Additionally, the models need to be scalable to accommodate potential changes in customer numbers over time. To address this, we propose implementing online learning, allowing the models to be trained periodically to adapt to evolving trends and customer behavior.

4. Scope and Objectives:

The scope of this project encompasses the development of predictive models for the specified milk products across four cities. Our objectives include minimizing prediction errors, maximizing the scalability of the models, and providing actionable insights for dairy industry stakeholders.

5. Data Description:

The dataset provided by Provilac Dairy includes historical records of milk consumption for 21 products across four cities. Variables such as date, quantity of milk sold, product type, and city location are included. Data preprocessing steps will include cleaning, normalization, and feature engineering to ensure the quality and relevance of the data.

6. Methodology:

Our proposed approach involves leveraging machine learning algorithms such as regression and time series analysis to develop the predictive models. To address scalability concerns, we plan to implement online learning, allowing the models to be trained periodically to adapt to changes in customer numbers and consumption patterns.

7. Results and Evaluation:

Upon implementation, the predictive models are expected to demonstrate high accuracy in estimating milk consumption for the specified products and cities. Evaluation of scalability will focus on the models' ability to adapt to changes in customer numbers and consumption patterns over time.

8. Discussion:

The findings of this project will provide valuable insights for dairy industry stakeholders, enabling them to make informed decisions regarding production, distribution, and sales strategies. The adoption of online learning techniques will ensure the scalability and adaptability of the predictive models in the dynamic dairy market.

9. Conclusion:

In conclusion, the development of predictive models for milk consumption prediction represents a significant step forward for the dairy industry. By leveraging historical data and advanced machine learning techniques, we can enhance decision-making processes and optimize resource allocation across the dairy supply chain.